

Ignacio Magaña Hernandez

Email: imhernan@andrew.cmu.edu
Web: <http://ignaciomagana.github.io>
Citizenship: USA

Address: Department of Physics
Carnegie Mellon University
5000 Forbes Ave
Pittsburgh, PA 15213

Employment

Carnegie Mellon University <i>Research Physicist</i>	Pittsburgh, PA 2026 – Present
Carnegie Mellon University <i>McWilliams Postdoctoral Fellow</i>	Pittsburgh, PA 2023 – 2026
University of Wisconsin-Milwaukee <i>NSF Graduate Research Fellow</i>	Milwaukee, WI 2017 – 2023
The Chinese University of Hong Kong <i>Undergraduate Research Assistant</i>	Hong Kong, Hong Kong 2016 – 2017
Caltech <i>LIGO SURF Fellow</i>	Pasadena, CA 2015

Education

University of Wisconsin-Milwaukee <i>PhD in Physics</i> Thesis: <i>Cosmology, Lensing, and Modified Gravity with Gravitational Waves</i> (Advisor: Patrick Brady)	Milwaukee, WI 2017 – 2023
University of California, Santa Barbara <i>B.S., Physics (Highest Honors)</i>	Santa Barbara, CA 2014 – 2017
Foothill College <i>Transfer Student, Physics</i>	Los Altos Hills, CA 2012 – 2014

Grants & Research Funding

• CMU McWilliams Seed Grant <i>Internal grant funding exploratory projects to generate preliminary results for high impact research.</i>	2025
• NSF ACCESS Allocation, Discovery <i>400K service units.</i>	2026
• NSF ACCESS Allocation PHY240308, Jetstream2 <i>Via the ACCESS program.</i>	2024
• Pittsburgh Supercomputing Center, Vera Cluster <i>Computing resources. Research supported by NSF Grant No. 2308193 (PI A. Palmese).</i>	

Awards & Fellowships

• CMU McWilliams Postdoctoral Fellowship <i>McWilliams Center for Cosmology and Astrophysics.</i>	2023 – 2026
• UWM Advanced Opportunity Program Fellowship <i>UWM fellowship supporting underrepresented graduate students for three years.</i>	2020 - 2023
• NSF Graduate Research Fellowship <i>NSF fellowship supporting graduate students for three years.</i>	2017 - 2022
• NASA Wisconsin Space Grant Consortium Fellowship <i>NASA funded fellowship for graduate research in STEM at Wisconsin institutions.</i>	2020 – 2022
• UWM Physics Department Lichtman Fellowship <i>Departmental fellowship awarded for exceptional academic achievement in physics.</i>	2020
• UWM Physics Department Lichtman Research Fellowship <i>Departmental research fellowship.</i>	
• UCSB Physics Department Highest Academic Honors	2017
• LIGO SURF Victor S. Blanco Fellowship <i>Joint LIGO SURF and National Society of Hispanic Physicists program supporting undergraduates in gravitational-wave science.</i>	2015
• NSF Science, Math and Engineering (S-STEM) Scholarship <i>NSF scholarship for STEM undergraduates.</i>	2013 – 2014

Publication Summary

h-index: 78 / 20 **Publications:** 133 / 37 (16 first/second-author) **Citations:** 64,168 / 2,411

First number includes LIGO collaboration papers; second excludes them. Counts from inspirehep.net, October 2026.

First/Second Author Publications

*Student-led under my supervision.

1. *Normalizing Flow Emulator for Gravitational Wave Detection Probability in the LIGO/Virgo/KAGRA O4 Observing Run*
J. Gassert*, **I. Magaña Hernandez**, A. J. Amsellem, A. Palmese and B. Póczos
([arXiv:2610.00306](https://arxiv.org/abs/2610.00306), 0 citations)
2. *Dark siren cosmology with the Dark Energy Spectroscopic Instrument*
I. Magaña Hernandez, A. Palmese et al.
Under DESI internal review
3. *Data-Driven Trends and Subpopulations in the Gravitational Wave Binary Black Hole Merger Population with UMAP*
A. J. Amsellem*, **I. Magaña Hernandez**, A. Palmese and J. Gassert
([arXiv:2603.06566](https://arxiv.org/abs/2603.06566), 1 citations)
4. *Lucky Strikes: On the Origins of GW190814 Through Isolated Binary Evolution*
I. Magaña Hernandez and K. Breivik
([arXiv:2511.16648](https://arxiv.org/abs/2511.16648), 1 citations) Submitted to *ApJL*
5. *Spectral siren cosmology from gravitational-wave observations in GWTC-4.0*
I. Magaña Hernandez and A. Palmese
([arXiv:2509.03607](https://arxiv.org/abs/2509.03607), 21 citations) Submitted to *Physical Review Letters*
6. *Astrophysics informed Gaussian processes for gravitational-wave populations: Evidence for the onset of the pair-instability supernova mass gap*
I. Magaña Hernandez and A. Palmese
([arXiv:2508.19208](https://arxiv.org/abs/2508.19208), 16 citations) Submitted to *Physical Review Letters*
7. *Evidence of a new feature in the binary black hole mass distribution at $70 M_{\odot}$ from gravitational-wave observations*
I. Magaña Hernandez and A. Palmese
Phys. Rev. D 111, 083031 (2025) ([arXiv:2407.02460](https://arxiv.org/abs/2407.02460), 21 citations)
8. *Searching for binary black hole sub-populations in gravitational wave data using binned Gaussian processes*
A. Ray, **I. Magaña Hernandez**, K. Breivik, J.D.E. Creighton
ApJ 991 17 (2025) ([arXiv:2404.03166](https://arxiv.org/abs/2404.03166), 52 citations)
9. *Beyond Gaps and Bumps: Spectral Siren Cosmology with Non-Parametric Population Models*
I. Magaña Hernandez and A. Ray
([arXiv:2404.02522](https://arxiv.org/abs/2404.02522), 34 citations) Submitted to *Physical Review D*
10. *On the association of GW190425 with its potential electromagnetic counterpart FRB 20190425A*
I. Magaña Hernandez, V. D'Emilio, S. Morisaki, M. Bhardwaj and A. Palmese
ApJL 971 L5 (2024) ([arXiv:2404.02496](https://arxiv.org/abs/2404.02496), 14 citations)
11. *Non-parametric inference of the population of compact binaries from gravitational wave observations using binned Gaussian processes*
A Ray, **I. Magaña Hernandez**, S. Mohite, J.D.E. Creighton and S. Kapadia
ApJ 957 37 (2023) ([arXiv:2304.08046](https://arxiv.org/abs/2304.08046), 92 citations)
12. *Measuring the polarization content of gravitational waves with strongly lensed binary black hole mergers*
I. Magaña Hernandez
([arXiv:2211.01272](https://arxiv.org/abs/2211.01272), 11 citations) Submitted to *Physical Review D*
13. *Constraining the number of spacetime dimensions from GWTC-3 binary black hole mergers*
I. Magaña Hernandez
Phys. Rev. D 107, 084033 (2023) ([arXiv:2112.07650](https://arxiv.org/abs/2112.07650), 23 citations)
14. *A Bayesian statistical framework for identifying strongly-lensed gravitational-wave signals*
R.K.L. Lo and **I. Magaña Hernandez**
Phys. Rev. D 107, 123015 (2023) ([arXiv:2104.09339](https://arxiv.org/abs/2104.09339), 102 citations)
15. *Identifying strong gravitational-wave lensing during the second observing run of Advanced LIGO and Advanced Virgo*
X. Liu, **I. Magaña Hernandez** and J.D.E. Creighton
ApJ 908 97 (2021) ([arXiv:2009.06539](https://arxiv.org/abs/2009.06539), 108 citations)

16. *Cosmological inference using gravitational wave standard sirens: A mock data analysis*
R. Gray, **I. Magaña Hernandez** et al.
Phys. Rev. D 101, 122001 (2020) ([arXiv:1908.06050](#), 274 citations)

Significant Contributions

1. *The multimessenger Universe as a training ground for frontier AI*
F. Petulante et al. (including **I. Magaña Hernandez**)
Nature Astron. 10, 938 (2026) ([10.1038/s41550-026-02910-w](#), 0 citations)
2. *One Year and One Night to 1% in H_0 : Efficient Spectroscopic Strategy for Dark Siren Cosmology*
Y. Dang, A. J. Amsellem, **I. Magaña Hernandez**, A. Palmese and B. S. Sathyaprakash
([arXiv:2607.17284](#), 0 citations)
3. *Electromagnetic Follow-up of the Sub-Solar Mass Gravitational Wave Candidate S251112cm: Kilonova Constraints and a Coincident Iib Supernova*
X. J. Hall et al. (including **I. Magaña Hernandez**)
([arXiv:2605.10940](#), 8 citations)
4. *Wide Jets or Low Rates: Reconciling Short GRB and Gravitational-Wave Neutron Star Merger Rates*
K. Kunnumkai, A. Palmese, B. O'Connor, A. Farah and **I. Magaña Hernandez**
([arXiv:2604.05046](#), 6 citations)
5. *Probing the environment around GW170817 with DESI: insights on galaxy group peculiar velocities for standard siren measurements*
A. J. Amsellem et al. (including **I. Magaña Hernandez**)
ApJ 1001, 157 (2026) ([arXiv:2512.08818](#), 5 citations)
6. *AT2025ulz and S250818k: Investigating early time observations of a subsolar mass gravitational-wave binary neutron star merger candidate*
X. Hall et al. (including **I. Magaña Hernandez**)
([arXiv:2510.24620](#), 18 citations)
7. *AT2025ulz and S250818k: Deep X-ray and radio limits on off-axis afterglow emission and prospects for future discovery*
B. O'Connor et al. (including **I. Magaña Hernandez**)
ApJL 995, L47 (2025) ([arXiv:2510.23728](#), 14 citations)
8. *AT2025ulz and S250818k: Leveraging DESI spectroscopy in the hunt for a kilonova associated with a sub-solar mass gravitational wave candidate*
X. Hall et al. (including **I. Magaña Hernandez**)
ApJL 1001, L20 (2026) ([arXiv:2510.23723](#), 14 citations)
9. *Kilonova Constraints for the LIGO/Virgo/KAGRA Neutron Star Merger Candidate S250206dm: GW-MMADS Observations*
L. Hu et al. (including **I. Magaña Hernandez**)
ApJL 990 L46 (2025) ([arXiv:2506.22626](#), 17 citations)
10. *A Dark Siren Measurement of the Hubble Constant with the LIGO/Virgo Gravitational Wave Event GW190412 and DESI Galaxies*
W. Ballard, A. Palmese, **I. Magaña Hernandez** et al.
Res. Notes AAS 7 250 (2023) ([arXiv:2311.13062](#), 35 citations)
11. *Follow-up analyses to the O3 LIGO–Virgo–KAGRA lensing searches*
J. Janquart et al. (including **I. Magaña Hernandez**)
MNRAS 526, 3832 (2023) ([arXiv:2306.03827](#), 83 citations)
12. *Challenges for Fast Radio Bursts as Multi-Messenger Sources from Binary Neutron Star Mergers*
M. Bhardwaj, A. Palmese, **I. Magaña Hernandez**, V. D’Emilio and S. Morisaki
ApJ 977 122 (2024) ([arXiv:2306.00948](#), 31 citations)
13. *Localization of binary neutron star mergers with a single Cosmic Explorer*
P. Baral, S. Morisaki, **I. Magaña Hernandez** and J.D.E. Creighton
Phys. Rev. D 108, 043010 (2023) ([arXiv:2304.09889](#), 17 citations)
14. *The Hitchhiker’s guide to the galaxy catalog approach for gravitational wave cosmology*
J. Gair et al. (including **I. Magaña Hernandez**)
AJ 166 22 (2023) ([arXiv:2212.08694](#), 119 citations)
15. *A joint ranking statistic for multi-messenger astronomical searches with gravitational waves*
B. Piotrkowski, A. Baylor and **I. Magaña Hernandez**
Class. Quantum Grav. 39 085010 (2022) ([arXiv:2111.12814](#), 8 citations)

16. *Rapid model comparison of equations of state from gravitational wave observation of binary neutron star coalescences*
S. Ghosh, X. Liu, J.D.E. Creighton, **I. Magaña Hernandez** et al.
Phys. Rev. D 104, 083003 (2021) ([arXiv:2104.08681](#), [25 citations](#))
17. *Gravitational wave friction in light of GW170817 and GW190521*
S. Mastroianni, L. Haegel, K. Karathanasis, **I. Magaña Hernandez** and D.A. Steer
JCAP 02 043 (2021) ([arXiv:2010.04047](#), [45 citations](#))
18. *Current observations are insufficient to confidently associate the binary black hole merger GW190521 with AGN J124942.3 + 344929*
G. Ashton, K. Ackley, **I. Magaña Hernandez** et al.
Class. Quantum Grav. 38 235004 (2021) ([arXiv:2009.12346](#), [114 citations](#))
19. *First measurement of the Hubble parameter from bright binary black hole GW190521*
S. Mukherjee et al. (including **I. Magaña Hernandez**)
([arXiv:2009.14199](#), [73 citations](#))
20. *Bayesian inference for compact binary coalescences with BILBY: Validation and application to the first LIGO–Virgo gravitational-wave transient catalogue*
I.M. Romero-Shaw et al. (including **I. Magaña Hernandez**)
MNRAS 499, 3295 (2020) ([arXiv:2006.00714](#), [642 citations](#))
21. *A statistical standard siren measurement of the Hubble Constant from the LIGO/Virgo gravitational wave compact object merger GW190814 and Dark Energy Survey galaxies*
A. Palmese et al. (including **I. Magaña Hernandez**)
ApJL 900 L33 (2020) ([arXiv:2006.14961](#), [174 citations](#))
22. *A standard siren measurement of the Hubble Constant from GW170817 without the electromagnetic counterpart*
M. Fishbach, R. Gray, **I. Magaña Hernandez** et al.
ApJL 871 L13 (2019) ([arXiv:1807.05667](#), [304 citations](#))

LIGO-Virgo Publications to which I highly contributed:

1. *GW190814: Gravitational Waves from the Coalescence of a 23 Solar Mass Black Hole with a 2.6 Solar Mass Compact Object*
LIGO Scientific and Virgo Collaborations (including **I. Magaña Hernandez**)
ApJL 896 L44 (2020) ([arXiv:2006.12611](#), [2121 citations](#)) **Led and wrote the Hubble constant analysis**
2. *A Gravitational-wave Measurement of the Hubble Constant Following the Second Observing Run of Advanced LIGO and Virgo*
LIGO Scientific and Virgo Collaborations (including **I. Magaña Hernandez**)
ApJ 909 218 (2021) ([arXiv:1908.06060](#), [424 citations](#)) **Paper writing and analysis team member**
3. *First measurement of the Hubble constant from a dark standard siren using the Dark Energy Survey galaxies and the LIGO/Virgo binary-black-hole merger GW170814*
M. Soares-Santos, A. Palmese et al. and LIGO Scientific and Virgo Collaborations (including **I. Magaña Hernandez**)
ApJL 876 L7 (2019) ([arXiv:1901.01540](#), [377 citations](#)) **Validated analysis using gwcsmo**

Invited Seminars

- | | |
|---|------|
| 1. American Museum of Natural History
<i>Binary black hole populations and cosmology with gravitational-wave observations and DESI</i> | 2025 |
| 2. Vanderbilt University
<i>Binary black hole populations and cosmology with gravitational-wave observations and DESI</i> | 2025 |
| 3. KAGRA Seminar
<i>Binary black hole populations and cosmology</i> | 2025 |
| 4. Stanford University (KIPAC Seminar)
<i>Binary black hole populations and cosmology with gravitational-wave observations and DESI</i> | 2025 |
| 5. Carnegie Mellon University (STAMPS Seminar)
<i>Binary black hole populations and cosmology</i> | 2025 |
| 6. Montclair State University
<i>Cosmology, Lensing, and Modified Gravity with Gravitational Waves</i> | 2023 |
| 7. Carnegie Mellon University
<i>Testing general relativity with strongly lensed gravitational waves</i> | 2023 |
| 8. Cardiff University
<i>Cosmology and modified gravity with binary black holes</i> | 2022 |

- | | |
|--|------|
| 9. UC Berkeley | 2022 |
| <i>Constraining the number of spacetime dimensions from GWTC-3 binary black hole mergers</i> | |
| 10. University of Wisconsin-Milwaukee | 2022 |
| <i>Cosmology and modified gravity with binary black hole mergers</i> | |
| 11. Eotvos Lorand University | 2019 |
| <i>Gravitational wave cosmology with large galaxy surveys</i> | |
| 12. University of Glasgow | 2019 |
| <i>Measuring the Hubble constant with gravitational-wave standard sirens</i> | |
| 13. The Chinese University of Hong Kong | 2019 |
| <i>A measurement of the Hubble constant from the second observation run of Advanced LIGO-Virgo</i> | |
| 14. Nagoya University | 2019 |
| <i>A measurement of the Hubble constant from the second observation run of Advanced LIGO-Virgo</i> | |

Contributed Talks

- | | |
|--|------|
| 1. Gravitational Waves Physics and Astronomy Workshop | 2025 |
| <i>Dark Siren Cosmology with the Dark Energy Spectroscopic Instrument (DESI)</i> | |
| 2. COSMO25 | 2025 |
| <i>Gravitational-wave dark siren cosmology with DESI</i> | |
| 3. Multimessenger Astronomy in the Era of Foundational AI | 2025 |
| <i>Trends in the GWTC-3 Binary Black Hole Merger Population with UMAP</i> | |
| 4. Summer DESI Collaboration Meeting | 2025 |
| <i>Gravitational-wave cosmology with DESI</i> | |
| 5. APS Global Physics Summit | 2025 |
| <i>Lucky Strikes: Unraveling the asymmetric origins of GW190814 through binary field evolution</i> | |
| 6. The 6th neighborhood workshop | 2024 |
| <i>Beyond Gaps and Bumps: Spectral Siren Cosmology with Non-Parametric Population Models</i> | |
| 7. Gravitational-wave populations: what's next? | 2023 |
| <i>Non-parametric inference of the population of compact binaries from gravitational wave observations using binned Gaussian processes</i> | |
| 8. APS April Meeting | 2023 |
| <i>Measuring the polarization content of gravitational waves with strongly lensed binary black hole mergers</i> | |
| 9. 11th Australasian Conference on General Relativity and Gravitation | 2022 |
| <i>Constraining the number of spacetime dimensions from GWTC-3 binary black hole mergers</i> | |
| 10. Midwest Relativity Meeting | 2021 |
| <i>Constraining the number of spacetime dimensions from GWTC-3 binary black hole mergers</i> | |
| 11. Midwest Relativity Meeting | 2020 |
| <i>On the candidate electromagnetic counterpart of GW190521 and its implications on the Hubble constant</i> | |
| 12. Gravitational Waves Physics and Astronomy Workshop | 2019 |
| <i>A gravitational-wave measurement of the Hubble constant following the second observing run of Advanced LIGO and Virgo</i> | |
| 13. Multi-Messenger Astrophysics in the Gravitational Wave Era | 2019 |
| <i>A gravitational-wave measurement of the Hubble constant following the second observing run of Advanced LIGO and Virgo</i> | |
| 14. YITP Asian-Pacific Winter School and Workshop on Gravitation and Cosmology | 2019 |
| <i>A standard siren measurement of the Hubble constant from GW170817 without the electromagnetic counterpart</i> | |
| 15. Midwest Relativity Meeting | 2018 |
| <i>Cosmological inference using gravitational wave standard sirens: A mock data challenge</i> | |
| 16. 19th Annual Physical Society of Hong Kong | 2016 |
| <i>Feedforward Seismic Noise Cancellation at the 40m Prototype Interferometer</i> | |

Other Workshops

- | | |
|---|------|
| 1. Multi-messenger Transients from Binary Mergers and Stellar Explosions | 2024 |
| <i>Aspen Center for Physics</i> | |
| 2. Black Holes and Neutron Stars with Gravitational Waves | 2019 |
| <i>Yukawa Institute for Theoretical Physics</i> | |

3. **The New Era of Gravitational-Wave Physics and Astrophysics** 2019
Kavli Institute for Theoretical Physics
4. **Asian-Pacific Winter School and Workshop on Gravitation and Cosmology** 2019
Yukawa Institute for Theoretical Physics
5. **Les Houches Summer School: Gravitational-Wave Astronomy** 2018
Ecole de Physique des Houches
6. **Summer School on Gravitational-Wave Astronomy** 2016
ICTS-TIFR

Service

- Referee: *The Astrophysical Journal*, *The Astrophysical Journal Letters*, *Physical Review D*, *Physical Review Letters*, *Nature Astronomy*
- Session Chair, Parallel 8a “GW/MMA II: theory,” COSMO-25 (2025)

Societies/Affiliations

- Dark Energy Spectroscopic Instrument (DESI) member (2023 – Present)
- LIGO Scientific Collaboration member (2015 – 2023, 2027 – Present)
- American Physical Society
- Society for Advancement of Chicanos/Hispanics and Native Americans in Science (SACNAS)
- National Society of Hispanic Physicists
- GW-MMADS O4 follow-up campaign collaborator

Teaching & Mentoring

- Teaching Assistant, General Relativity, University of Wisconsin-Milwaukee (2019)
Student Mentoring:
- **Kaitlyn Pak** Carnegie Mellon University
Undergraduate student - Multi-messenger associations with GWs 2025 – Present
- **Ariel Amsellem** Carnegie Mellon University
PhD student - Standard siren cosmology. 2023 – Present
- **Simon Yeung** University of Wisconsin-Milwaukee
PhD student - GW population systematics. 2023 – Present
- **Julia Yoon** Carnegie Mellon University
Undergraduate student - Dark siren cosmology with GW170817 2024 – 2025
- **Diego Michel** Carnegie Mellon University
Undergraduate student - Gaussian processes for GW populations. 2023 – 2024
- **Anarya Ray** University of Wisconsin-Milwaukee
PhD student (co-advised) - Binned Gaussian processes for GW populations. 2021 – 2024
- **Pratyusava Baral** University of Wisconsin-Milwaukee
PhD student (co-advised) - Localization of BNS events with 3G GW detectors. 2021 – 2023
- **Brandon Piotrkowski** University of Wisconsin-Milwaukee
PhD student (co-advised) - Multi-messenger association statistics. 2019 – 2020

Software

- **darksirens** — Hierarchical population and cosmological inference for gravitational-wave sirens.
<https://github.com/ignaciomagana/darksirens-core>
- **gwcats** — Preprocessing of GWTC PE files and LVK injection sets for darksirens.
<https://github.com/ignaciomagana/gwcats>
- **tinyns** — Tiny dynesty-style nested sampler for JAX-friendly likelihoods.
<https://github.com/ignaciomagana/tinyns>

Skills

- **Programming skills:**
 - Proficiency with: Python (NumPy, SciPy, Pandas, Scikit-learn), JAX, C, Mathematica, LaTeX.
 - Familiar with: Stan, C++, MATLAB/Simulink.
 - LVK related software: Bilby, LALSuite, PyCBC, ligo.skymap, gwcosmo (co-developer), hanabi.
- **Data Analysis:**
 - Bayesian model selection and parameter estimation, Monte Carlo methods.
 - Digital signal processing, machine learning, and basic numerical relativity experience.
- **Languages:** English (fluent), Spanish (fluent), and Korean (intermediate).

References

Antonella Palmese

Carnegie Mellon University
apalmese[at]andrew.cmu.edu

Katie Breivik

Carnegie Mellon University
kbreivik[at]andrew.cmu.edu

B.S. Sathyaprakash

Pennsylvania State University
bss25[at]psu.edu

Archisman Ghosh

Ghent University
archisman.ghosh[at]ugent.be

Soichiro Morisaki

The University of Tokyo
soichiro[at]icrr.u-tokyo.ac.jp

Kipp Cannon

The University of Tokyo
kipp[at]resceu.s.u-tokyo.ac.jp